Neural ODE based modeling of memristive devices

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***Abstract*—The development of modern memristive technologies requires accurate dynamic models to enable reliable simulation and prototyping in applications such as resistive RAM, neural networks, and neuromorphic computing. Existing symbolic models often exhibit limitations in accurately capturing the complex dynamic behavior of physical devices across various operating conditions, impacting their reliability in simulations. This work proposes a neural-network-based modeling of real memristors behaviour using Neural Ordinary Differential Equation (Neural ODE) models. Prior neural-network-based studies were limited by training data derived solely from theoretical models, preventing reliable assessment on real devices. The proposed method leverages a backpropagation technique based on solving the corresponding adjoint ODE backward in time using adjoint sensitivity methods. Experimental data obtained directly from Self-Directed Channel (SDC) memristors are used to train and validate the model. Results show that the Neural ODE approach more accurately reproduces the device's dynamic characteristics and provides improved flexibility and robustness compared to traditional theoretical models. These findings demonstrate the feasibility of Neural ODE–based frameworks for data-driven modeling of a broad class of physical memristive devices.**

***Index Terms*—Memristor modeling, artificial neural network, neural ODE.**

I. Introduction

NUMEROUS memristor models have been proposed following the discovery of memristive behavior at the nanoscale [1]. Physics-based approaches range from generalized device models capturing threshold-dependent switching [2], [3], [4] to compact representations of RRAM structures such as the Stanford model [5]. For Self-Directed Channel (SDC) memristors specifically, the Generalized Metastable Switch (GMS) model [6] and its derivatives—the Mean Metastable Switch (MMS) [7] and Generalized MMS (GMMS) [8]—provide semi-empirical formulations based on transition probabilities between metastable states.

Despite their widespread adoption, these symbolic models share common limitations: they require extensive parameter tuning, employ window functions that introduce numerical discontinuities [9], and exhibit limited generalization when operating conditions deviate from calibration regimes [10]. A recent attempt to address these issues using Physics-Informed Neural Networks [11] demonstrated the potential of data-driven approaches, but relied solely on synthetic data generated from existing theoretical models rather than physical device measurements.

In [12], the authors introduced Neural Ordinary Differential Equations (Neural ODEs), where hidden-state dynamics are governed by an ODE with a neural network defining the vector field. Training is performed end-to-end using adjoint sensitivity methods for efficient gradient computation.

This paper proposes two Neural ODE architectures for modeling memristive devices trained directly on experimental data from SDC memristors. The first, **SingleNN-MemODE**, uses a physics-informed memductance with a neural network modeling only the state dynamics. The second, **DualNN-MemODE**, employs neural networks for both memductance and state dynamics, enabling a higher-dimensional latent representation. We demonstrate that both models outperform the MMS and GMMS benchmarks in prediction accuracy while maintaining comparable or shorter training times.

The paper is organized as follows. Sec. II presents the proposed architectures, the objective function, and the reference models. Sec. III reports experimental results and comparative analysis. The last section concludes the study.

The remaining part of this paper is organized as follows. In Sec. II, the neural network architecture is presented and two neural network-based models (SingleNN-MemODE and DualNN-MemODE) are proposed. The objective function defining the optimization problem to be solved is defined and existing memristors models used for comparison are briefly described. In Sec. III, proposed models are trained using experimental data to obtain accurate models of real SDC memristors. The results are compared with the results obtained using existing methods. The influence of hyperparameters of the proposed methods on their performance is discussed. The last section concludes the study.

II. Modeling Memristors using Neural Networks

This study utilizes experimental data obtained from Self-Directed Channel (SDC) memristors doped with tungsten. The structure and properties of these devices have been described in detail in the literature, e.g., in [13], [14].

The experimental setup, illustrated in Fig. 1, consists of an SDC memristor connected in series with a resistor. The sinusoidal input voltage is supplied by an arbitrary waveform generator. Voltage signals are measured using a data acquisition (DAQ) system. Measurements are carried out for the following combinations of supply voltage amplitudes and frequencies: $V_s \in \{0.5,\ 1.0,\ 1.5\}$ [V], $f \in \{1,\ 5,\ 10,\ 20,\ 50,\ 100\}$ [Hz].

The analyzed model of a memristor follows the general theoretical framework of memristive devices originally introduced in [15], and is expressed as

$$i(t) = \mathcal{G}\left(\mathbf{x}(t), v(t)\right) v(t), \tag{1a}$$

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f\left(\mathbf{x}(t), v(t)\right), \tag{1b}$$

where $\mathbf{x}$ denotes the vector of internal state variables, whose temporal evolution is governed by the function f, and $\mathcal{G}$

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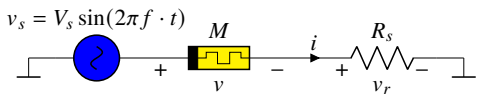

Fig. 1. Schematic diagram of the measurement setup used for the SDC memristor. The device is connected in series with a resistor, and the input voltage is applied via an arbitrary waveform generator. Voltage measurements are acquired using a data acquisition (DAQ) system.

represents the memductance of the device. The fundamental challenge in memristor modeling lies in accurately identifying the functional forms of $\mathcal{G}(\mathbf{x}, v)$ and $f(\mathbf{x}, v)$.

A. Objective Function

The objective function $\mathcal{L}$ to be minimized in the optimization process is based on trajectories generated by the dynamical system. It is designed to simultaneously account for both the signal values and its dynamic structure. To this end, it is defined as the sum of four components, each serving a distinct and complementary role in the optimization process

$$\mathcal{L} = \lambda_{\text{base}}\mathcal{L}_{\text{b}} + \lambda_{\text{vel}}\mathcal{L}_{\text{v}} + \lambda_{\text{curv}}\mathcal{L}_{\text{c}} + \sum_{i=1}^{N_c} \lambda_{\text{con}_i}\mathcal{L}_{\text{con}_i}(x). \quad (2)$$

The first term $\mathcal{L}_{\text{b}}$ denotes the primary error computed as the mean squared error (MSE) between predicted and reference values. $\mathcal{L}_{\text{v}}$ enforces accurate modeling of derivatives of state variables, $\mathcal{L}_{\text{c}}$ promotes consistency of the trajectories in terms of their curvature by computing MSE of second derivatives, and $\mathcal{L}_{\text{con}}(x)$ enforces fulfillment of physical or structural constraints that the model is required to satisfy, including conformity to the underlying physical model. The weighting coefficients λ_{base}, λ_{vel}, and λ_{curv} balance the influence of each term according to the priorities of the modeling task.Their values are calculated during each optimization step using the Geometric Loss Strategy (GLS) [16], [17].

Let us consider two sequences $z_k = (z_{k,n})_{n=1}^{N}$ and $\tilde{z}_k = (\tilde{z}_{k,n})_{N=1}^{N}$, where $z \in \{v, i, \dot{v}, \dot{i}, \ddot{v}, \ddot{i}\}$ is a selected variable (voltage or current) or its derivative obtained from measurements, $\tilde{z}$ is the predicted value of z, N is the number of samples and $k = 1, 2, \ldots, N_{\text{traj}}$ is the sequence index. The sequences are normalized utilizing the standard deviation to ensure statistical consistency within each dataset.

By implementing trajectory-specific normalization rather than global standardization, the methodology preserves the inherent temporal dynamics and statistical properties unique to each circuit configuration while simultaneously ensuring that amplitude disparities do not introduce systematic bias during predictive modeling procedures. This approach is particularly advantageous in scenarios involving varying operational conditions, where maintaining the relative temporal structure is paramount for accurate system identification and dynamic analysis. The normalized mean squared error between sequences $(z_{k,n})_{n=1}^{N}$ and $(\tilde{z}_{k,n})_{n=1}^{N}$ is defined as

$$\text{NMSE}(z_k, \tilde{z}_k) = \text{MSE}\left(\frac{z_k}{\sigma_{z,k}}, \frac{\tilde{z}_k}{\sigma_{z,k}}\right) = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{z_{k,n}}{\sigma_{z,k}} - \frac{\tilde{z}_{k,n}}{\sigma_{z,k}}\right)^2. \quad (3)$$

The primary component of the loss function is responsible for model fitting to experimental voltage-current data:

$$\mathcal{L}_{\text{b}} = \frac{1}{N_{\text{traj}}}\sum_{k=1}^{N_{\text{traj}}}\left(\text{NMSE}(v_k, \tilde{v}_k) + \text{NMSE}(i_k, \tilde{i}_k)\right), \quad (4)$$

where $\tilde{v}_k$ and $\tilde{i}_k$ represent the predicted voltage and current for the kth trajectory, while v_k and i_k denote the corresponding experimental measurements.

The first-order dynamics comparison $\mathcal{L}_{\text{v}}$ enables evaluation of temporal derivative agreement between model predictions and experimental data, particularly critical for capturing rapid memristor state transitions

$$\mathcal{L}_{\text{v}} = \frac{1}{N_{\text{traj}}}\sum_{k=1}^{N_{\text{traj}}}\left(\text{NMSE}(\dot{v}_k, \tilde{\dot{v}}_k) + \text{NMSE}(\dot{i}_k, \tilde{\dot{i}}_k)\right). \quad (5)$$

The second-order comparison term $\mathcal{L}_{\text{c}}$ addresses trajectory curvature characteristics

$$\mathcal{L}_{\text{c}} = \frac{1}{N_{\text{traj}}}\sum_{k=1}^{N_{\text{traj}}}\left(\text{NMSE}(\ddot{v}_k, \tilde{\ddot{v}}_k) + \text{NMSE}(\ddot{i}_k, \tilde{\ddot{i}}_k)\right). \quad (6)$$

This component facilitates superior representation of curvature properties within the phase-space trajectory, particularly important for modeling systems exhibiting complex nonlinear and highly dynamic responses.

To enforce parameter bounds within specified intervals, a sophisticated constraint loss function utilizing smooth sigmoid transitions is implemented. This component prevents parameter drift beyond physically meaningful ranges while maintaining differentiability for gradient-based optimization algorithms. The constraint loss for the admissibility interval $x \in [a, b]$ is expressed as:

$$\mathcal{L}_{\text{con}}(x) = \mathbb{E}\left[\sigma\left(\frac{a-x}{\gamma}\right)(a-x)^2 + \sigma\left(\frac{x-b}{\gamma}\right)(x-b)^2\right], \quad (7)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the sigmoid function, a and b denote the lower and upper bounds of the admissible parameter range, γ is a scaling factor controlling the steepness of the transition near the bounds, and $\mathbb{E}[\cdot]$ denotes the expected value (ensemble average over samples or trajectories). The term $\sigma((a-x)/\gamma)(a-x)^2$ penalizes values of x below the lower bound a, while the term $\sigma((x-b)/\gamma)(x-b)^2$ penalizes values of x above the upper bound b. Squaring ensures that the penalty is non-negative and grows quadratically with the violation magnitude.

B. Neural ODE based Memristor Modeling

In this study, two neural network-based modeling approaches are proposed called. The first one, denoted as **SingleNN-MemODE**, employs a deterministic formulation of the memristor memductance $\mathcal{G}$, while the function f defining the dynamics of internal variables is modeled using a neural network. In the second one, referred to as **DualNN-MemODE**, both $\mathcal{G}$ and f are modeled using neural networks. The conceptual architecture of the proposed approaches is illustrated in Fig. 2.

For the SingleNN-MemODE approach, the function f is implemented using an artificial neural network with a single

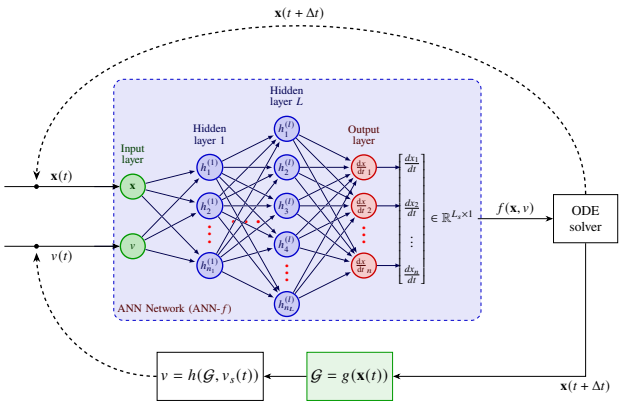

Fig. 2. The MemODE approach. The artificial neural network (ANN) evaluates the function $f(x, v)$, which is then integrated by the ODE solver to predict the state at the next time step $x(t + \Delta t)$. The upper dashed feedback loop indicates the recurrent nature of the process, where the output state is fed back as input for subsequent predictions. The memductance $\mathcal{G}$ is defined as function g which can be a deterministic function based on the internal variable x (as in SingleNN-MemODE) or the neural network with input $\mathbf{x}(t)$ (as in DualNN-MemODE). The function $h(\mathcal{G}, v_s(t))$ computes the memristor voltage v given the memductance $\mathcal{G}$ and supply voltage $v_s(t)$.

output. For the DualNN-MemODE approach the number of outputs is equal to the number of internal variables and may be larger than 1.

In the SingleNN-MemODE approach, the memductance is defined on the basis of the physical mechanisms underlying self-directed channel (SDC) devices, which is expressed as a function of the internal state variable x as follows:

$$\mathcal{G}(x) = \frac{x}{R_{\text{ON}}} + \frac{1-x}{R_{\text{OFF}}}, \tag{8}$$

where R_{ON} and R_{OFF} denote the resistances in the low-resistance and high-resistance states, respectively. During the optimization process, the network parameters (weights $w_{k,i}$ and biases b_k) are tuned jointly with the values of R_{ON} and R_{OFF}. For this approach the following constraints are applied:

- $R_{\text{ON}} \in (500\,\Omega, R_{\text{OFF}})$,
- $R_{\text{OFF}} \in (R_{\text{ON}}, 200\,\text{k}\Omega)$,
- $x(t) \in (0, 1)$.

The lower bound for R_{ON} and the upper bound for R_{OFF} are based on empirical measurements. Remaining constraints are consequences of physically meaningful values of parameters and variables.

In the DualNN-MemODEapproach, the memductance is modeled using the second neural network. This neural network takes the current state $\mathbf{x}$ as the input and produces the corresponding memductance $\mathcal{G}(\mathbf{x})$ as the output. The memristor state $\mathbf{x}$ is not constrained to be one-dimensional, enabling a more flexible representation of the device's internal dynamics. The state variables are not restricted to any specific range, and hence no explicit constraints are applied to them. For this approach the only constraint is

- $\mathcal{G}(\mathbf{x}) \in (5\,\mu\text{S}, 2\,\text{mS})$.

These bounds are based on measurements.

The learning process involves iteratively adjusting the network parameters to minimize the loss function that quantifies the difference between predicted and target outputs. This is achieved through backpropagation algorithm combined with the gradient descent optimization, where gradients of the loss function with respect to each parameter are computed and used to update the weights and biases in the direction that reduces the overall error.

C. Neural Network Architecture

The neural network training is implemented using the `JAX` library [18] for array-oriented numerical computation and `Equinox` [19] for neural network construction in `Python`. The parameter gradients of the ODE solution are computed using adjoint sensitivity methods [20], [12]. The `tsit5` adaptive Runge-Kutta solver [21] is employed for its computational efficiency and numerical accuracy, with adaptive step-size control ensuring stable integration during training.

The neural network architecture, illustrated in Fig. 3, employs an expansion–compression (diamond-shaped) topology that incorporates a double reduction in dimensionality in both the initial and final hidden layers. This architectural configuration substantially decreases the overall number of trainable parameters and helps to mitigate vanishing and exploding gradient issues commonly encountered in deep architectures processing temporal sequences.

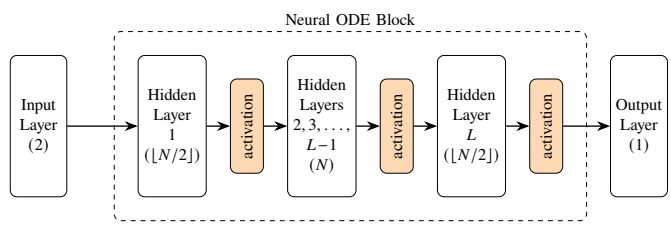

Fig. 3. Architecture of the feedforward neural network used within the Neural ODE framework for memristor dynamics modeling. The network consists of an input layer receiving two inputs, multiple hidden layers with decreasing-increasing-decreasing neuron counts $(\lfloor N/2 \rfloor, N, \ldots, N, \lfloor N/2 \rfloor)$, activation functions between layers, and a single output. The Neural ODE block encompasses the core computational layers that approximate the function $f(x(t), v(t))$ of the ODE system.

Several techniques are tested for the optimization of network weights and the `Adam` (Adaptive Moment Estimation) optimizer is selected due to its superior performance. The choice of an optimizer and its parameters is guided by hyperparameter tuning conducted with the `Optuna` framework [22]. The hyperparameter search space comprises the learning rate, weight decay, batch size, and architecture parameters like the width and the depth of the neural network. The optimal configuration is selected based on minimizing the objective function on the validation set, ensuring that the model generalizes well to unseen data.

D. Existing Memristor Models for Comparison

To evaluate the effectiveness of the proposed models, their performance is compared with the performance of two existing memristor models: the Mean Metastable Switch (MMS) model [7] and the Generalized Mean Metastable Switch Memristor (GMMS) model [6], [8].

The MMS model is a simplified representation of memristor's behavior, focusing on the average switching characteristics rather than the detailed dynamics. It captures essential features of the device by describing transitions between the high-resistance state (HRS) and the low-resistance state (LRS) based on the applied voltage. This model describes memristor's dynamics as

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{\tau}\left(f_{\mathrm{ON}}(v)(1-x) - f_{\mathrm{OFF}}(v)x\right), \tag{9}$$

$$f_{\mathrm{ON}}(v) = \sigma\left(-\beta(v - V_{\mathrm{ON}})\right), \tag{10}$$

$$f_{\mathrm{OFF}}(v) = 1 - \sigma\left(\beta(v - V_{\mathrm{OFF}})\right), \tag{11}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$, $\beta = \frac{q}{kT}$, k is the Boltzmann constant, T is the absolute temperature, q represents the elementary charge, while V_{ON} and V_{OFF} denote the switching voltages to LRS and HRS states, respectively. The instantaneous memductance is defined in (8).

In the GMMS model memristors are represented as a parallel connection of memory-dependent element and a Schottky diode [6], [8]. This formulation extends the original metastable switch concept by incorporating nonlinear current–voltage characteristics, enabling a more accurate representation of memristor's behavior across a wide range of operating conditions. The evolution of the state variable is governed by (9), while the current–voltage relationship is

$$\begin{aligned} i(t) &= \eta i_{\mathrm{md}}(v,x) + (1-\eta) i_{\mathrm{d}}(v), \\ i_{\mathrm{d}}(v) &= \alpha_{\mathrm{f}} e^{\beta_{\mathrm{f}} v} - \alpha_{\mathrm{r}} e^{-\beta_{\mathrm{r}} v}, \\ i_{\mathrm{md}}(v,x) &= \mathcal{G}(x) v, \end{aligned} \tag{12}$$

where $\eta \in [0,1]$ is the parameter representing the ratio of the memory dependent current i_{md} and total current i flowing through the device. The parameters α_f, β_f, α_r, and β_r are positive constants characterizing the forward and reverse current across the Schottky barrier.

The inclusion of additional parameters and nonlinearities in the GMMS model increases model's complexity and poses challenges for parameter optimization. An additional challenge arises due to the Schottky effect, which introduces nonlinearity into the current–voltage characteristic. When the memristor is connected in series with a resistor, this requires solving a set of differential-algebraic equations (DAEs) of the form

$$\begin{cases} v_{\mathrm{s}} = v_{\mathrm{r}} + v, \\ \dfrac{\mathrm{d}x}{\mathrm{d}t} = \dfrac{1}{\tau}\left(f_{\mathrm{ON}}(v)(1-x) - f_{\mathrm{OFF}}(v)x\right), \end{cases} \tag{13}$$

where v_{s} denotes the supply voltage, $v_{\mathrm{r}} = R_{\mathrm{s}} i$ is the voltage across the resistor, and v is the voltage across the memristor.

Parameter estimation for existing memristor models employs a two-stage hybrid optimization strategy. Global exploration is performed using Adaptive Differential Evolution with radius-limited sampling from the `BlackBoxOptim` package [23] in `Julia`, followed by local refinement via the L-BFGS-B quasi-Newton method [24] implemented in `Optim.jl`. This approach efficiently identifies optimal parameter sets while avoiding premature convergence to local minima.

III. Results

During the hyperparameter tuning, a range of neural network architectures is systematically evaluated in order to identify the most suitable configuration for modeling memristor dynamics. The evaluation considers the number of hidden layers, the number of neurons per layer, and optimization settings (e.g. the batch size and the learning rate), with the aim of achieving a balance between model accuracy and generalization capabilities, as described in Sec. II-C. Among these settings, the batch size determines the number of training samples processed before the model parameters are updated, thereby influencing both the convergence speed and the stability of the learning process. In this study, the batch size corresponds to the number of trajectories processed in parallel before each parameter update. The learning rate, on the other hand, controls the step size of each parameter update during training. An excessively high learning rate may cause the optimization process to diverge, while a low value may result in slow convergence or being trapped in a local minimum.

In this study, the dataset consists of 18 trajectories obtained from measurements for all combinations of supply voltage amplitudes $V_{\mathrm{s}} \in \{0.5\,\mathrm{V}, 1.0\,\mathrm{V}, 1.5\,\mathrm{V}\}$ and frequencies $f \in \{1\,\mathrm{Hz}, 5\,\mathrm{Hz}, 10\,\mathrm{Hz}, 20\,\mathrm{Hz}, 50\,\mathrm{Hz}, 100\,\mathrm{Hz}\}$. The number of samples for each trajectory is 10^5 (100 periods of the supply voltage per trajectory, 1000 samples per period). This dataset is divided into the training dataset consisting of 14 trajectories and the validation datasets comprising 4 trajectories. Trajectories used for training and validation are selected to cover a diverse range of memristor's operating conditions, thereby enhancing the model's ability to generalize across different dynamic regimes. The validation dataset contains the following pairs of supply voltages and frequencies: $(1.5\,\mathrm{V}, 5\,\mathrm{Hz})$, $(1\,\mathrm{V}, 1\,\mathrm{Hz})$, $(1\,\mathrm{V}, 100\,\mathrm{Hz})$, and $(0.5\,\mathrm{V}, 5\,\mathrm{Hz})$.

During the hyperparameter optimization stage, the optimal configuration of the SingleNN-MemODE architecture found comprises four hidden layers with 64, 128, 128, and 64 neurons in consecutive layers. The model is trained using the `AdamW` optimizer with a learning rate of approximately 1.89×10^{-2} and a batch size of 2. The swish activation function $\varphi(x) = x\cdot\sigma(x)$ is employed for the hidden layers, while the final layer utilizes the linear activation function $\varphi(x) = x$.

For the DualNN-MemODE architecture, the optimal neural network configuration responsible for computing $f(\mathbf{x}, v)$ consists of four hidden layers with 120, 240, 240, and 120 neurons in consecutive layers and with 6 latent states, corresponding to the dimensionality of the internal state vector $\mathbf{x}$. The auxiliary neural network responsible for computing $\mathcal{G}$ includes one hidden layer containing 80 neurons. Both networks are trained using the `AdaBelief` optimizer with a learning rate of approximately 6.30×10^{-3} and a batch size of 14. The first neural network employs the GELU activation function $\varphi(x) = x \cdot \Phi(x) \approx 0.5x(1 + \tanh[\sqrt{2/\pi}(x + 0.044715x^3)])$ in the hidden layers and the linear activation in the output layer, whereas the second network uses the ReLU activation function $\varphi(x) = \max(0, x)$ with the sigmoid output activation.

A representative learning curve corresponding to the SingleNN-MemODE architecture, is presented in Fig. 4. The

training process is conducted over 360 epochs, with the objective function $\mathcal{L}$ monitored on both the training and validation datasets to provide insights into convergence behavior and potential overfitting. The learning curve demonstrates a consistent decrease in values of $\mathcal{L}$ for both datasets, which indicates effective optimization and a satisfactory model fitting to the experimental data. The close agreement between the values of the objective functions for the training and validation datasets further confirms the generalization ability of the selected architecture.

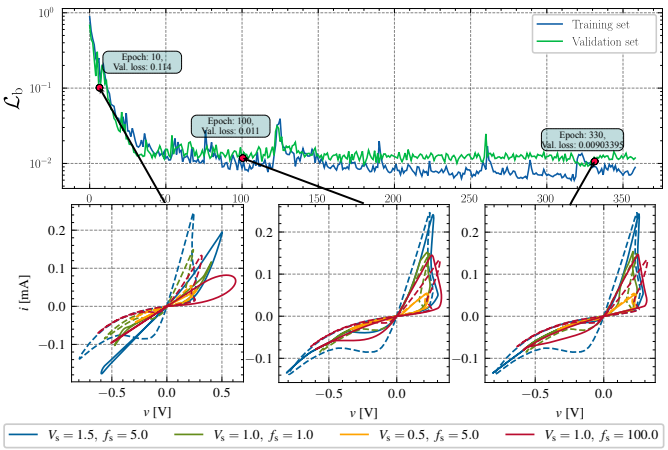

Fig. 4. Learning curve for the SingleNN-MemODE neural network architecture during training over 360 epochs. The insets illustrate v–i hysteresis loops at selected epochs, where the dashed lines correspond to the measured responses and the solid lines denote the model predictions.

Fig. 4 also presents the v–i hysteresis characteristics obtained at selected stages of the optimization process. These snapshots highlight how the model progressively refines its internal representation of the device dynamics during training. The gradual improvement of the hysteresis fit across epochs illustrates the capability of the neural network to capture the nonlinear and history-dependent behavior of the memristor, thereby validating the suitability of the chosen architecture and training configuration.

The effectiveness of optimizing the full loss function $\mathcal{L}$ defined in (2) was compared against optimizing only the base loss function $\mathcal{L}_b$, defined in (4). For instance, when optimizing only the base loss function for the SingleNN-MemODE architecture, the final value obtained for the validation dataset is $\mathcal{L}_b = 2.24\times10^{-2}$, whereas optimizing the full loss function yields a lower value of $\mathcal{L}_b = 1.85\times10^{-2}$. This result indicates that incorporating additional terms $\mathcal{L}_v$ and $\mathcal{L}_c$ improves convergence and enhances the overall predictive performance of the model, as reflected by reduced validation loss values and higher accuracy on unseen data.

Representative simulation results demonstrating the neural network model's predictive capabilities are presented in Fig. 5 through comparative analysis of a tungsten-doped memristor device subjected to sinusoidal voltage excitation with an amplitude of 1.5 V and frequency of 100 Hz.

The analysis encompasses the voltage waveform, the current waveform and the voltage-current hysteresis relationships. The voltage profile confirms the fidelity of the input stimulus. The

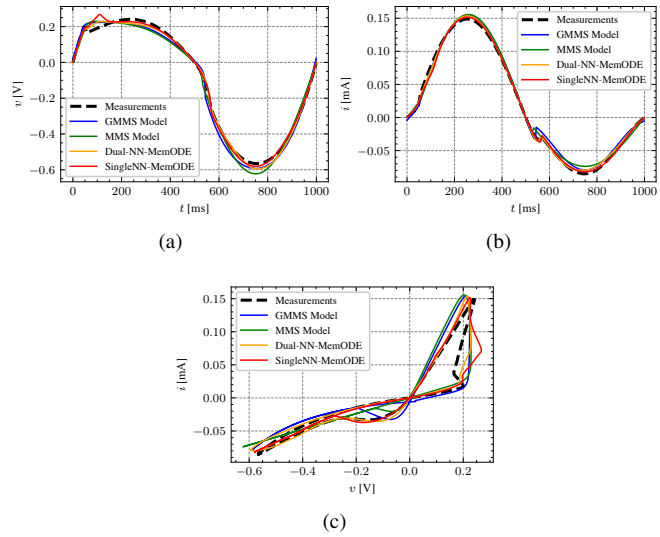

Fig. 5. Simulation test results for tungsten-doped memristor dynamics under sinusoidal voltage excitation (amplitude: 1.0 V, frequency: 1.0 Hz); (a) memristor's voltage waveform, (b) memristor's current response, (c) voltage-current hysteresis loop ($v - i$) illustrating the characteristic pinched behavior and memory properties fundamental to memristive operation.

temporal current response reveals the device's instantaneous electrical behavior and switching kinetics. The voltage-current hysteresis loops ($v - i$) illustrate the fundamental memory properties that define memristive behavior.

The hysteresis loops are particularly diagnostic of memristor performance, as their shape, area, and switching thresholds directly reflect the underlying ionic transport mechanisms and structural modifications responsible for resistive switching. The pinched hysteresis loop, which intersects at the origin, serves as a definitive signature of memristive behavior, while the loop area quantifies the energy dissipation associated with switching events. These features enable comprehensive validation of the neural network model's ability to capture both the static and dynamic aspects of memristor operation.

A. Comparative Analysis with the MMS and GMMS Models

To assess the performance of the neural ODE approach, a comprehensive comparative analysis is conducted between the developed neural network models and the existing models of SDC memristors: the Mean Metastable Switch (MMS) model and the Generalized Mean Metastable Switch (GMMS) model.

This comparison is particularly significant as the MMS and GMMS models have been extensively validated against experimental data and serve as benchmarks for memristor circuit simulation [8], [25]. The comparative evaluation encompasses temporal voltage and current evolution, hysteretic characteristics, and quantitative loss function metrics to provide comprehensive assessment of modeling fidelity.

Comparison results are plotted in Fig. 6. Quantitative analysis reveals substantial performance improvements achieved by the neural network approach.

Table **??** presents a comparative analysis of the loss functions $\mathcal{L}_b$ obtained for the proposed and existing models, offering a quantitative assessment of their performance. The

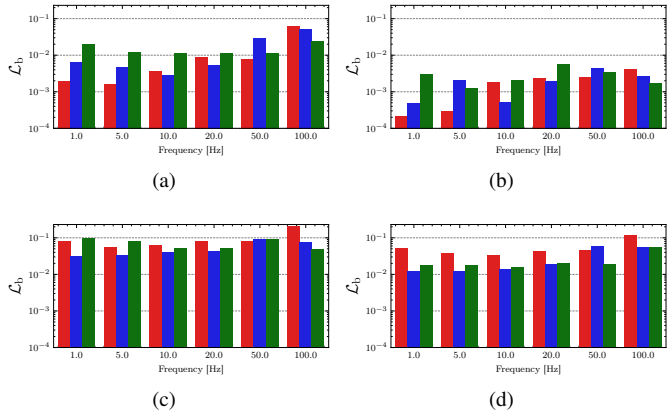

Fig. 6. Comparative analysis of the loss function $\mathcal{L}_b$: (a) the SingleNN-MemODE model, (b) the DualNN-MemODE model, (c) the MMS model, (d) the GMMS model. Results for voltage amplitudes $V_s = 0.5, 1.0, 1.5$ V are plotted in red (■), blue (■) and green (■), respectively.

TABLE I
COMPARATIVE ANALYSIS OF THE LOSS FUNCTION $\mathcal{L}_b$ FOR THE VALIDATION, TRAINING AND THE WHOLE DATASETS.

	SingleNN-MemODE	DualNN-MemODE	GMMS Model	MMS Model
Validation dataset				
$\mathcal{L}_b$	1.73×10^{-2}	5.77×10^{-3}	2.74×10^{-2}	2.51×10^{-2}
Training dataset				
$\mathcal{L}_b$	1.46×10^{-2}	5.11×10^{-3}	3.42×10^{-2}	4.22×10^{-2}
Whole dataset				
$\mathcal{L}_b$	1.52×10^{-2}	5.26×10^{-3}	3.27×10^{-2}	3.84×10^{-2}

results highlight the neural network's enhanced capability to reproduce complex and subtle aspects of memristor dynamics that are insufficiently captured by the phenomenological MMS/GMMS framework. For clarity, the highest loss value in each row is marked in red, whereas the lowest is marked in green. Overall, the neural network model consistently achieves substantially lower loss values, demonstrating superior accuracy and fidelity in representing the device behavior.

Table I reports the average loss function values $\mathcal{L}_b$ calculated for the validation, training and the whole datasets for the proposed and existing models. Results obtained for the validation dataset provide a quantitative evaluation of generalization performance of tested models across unseen memristor dynamics. The best result of 5.77×10^{-3} is obtained for the DualNN-MemODE model. The significant reduction of the average values of $\mathcal{L}_b$ for both proposed method when compared to existing methods demonstrates the enhanced modeling capability achieved through the neural ODE framework.

The superior performance of the neural network model can be attributed to its capacity for learning complex nonlinear mappings directly from experimental data, whereas the MMS model relies on predetermined phenomenological relationships that may not fully capture the device-specific dynamics and material-dependent switching characteristics inherent in real memristor devices.

Let us now compare computation times needed to find different models and evaluation times of models. Computations are carried out using a 3.5 GHz Intel i9-9900X processor. The median training duration for the SingleNN-MemODE and DualNN-MemODE architectures is approximately 11.3 minutes and 13.8 minutes, respectively. The median computational time necessary for parameter optimization of the MMS model is approximately 20.4 minutes, whereas the GMMS model requires approximately 8.4 minutes using the same hardware. The neural network architectures are trained employing a standard deep learning framework based on gradient descent optimization algorithms. Conversely, the parameter identification methodology for the analytical models (MMS and GMMS) requires a two-stage optimization procedure. Initially, a computationally intensive global search algorithm is executed over the whole parameter space to obtain an approximate yet robust initial parameter estimate. Subsequently, a local refinement procedure employing gradient-based optimization is applied. This methodology reflects the inherent computational complexity of determining appropriate initial parameters for analytical models and accounts for their extended calibration duration relative to the single-iteration training cost of neural network-based architectures.

The computation time for trajectory generation comprising 18 trajectories using the MMS model is approximately 0.047 seconds, whereas the GMMS model requires approximately 0.051 seconds. In contrast, the DualNN-MemODE architecture needs approximately 3.97 seconds, while the SingleNN-MemODE architecture requires approximately 5.56 seconds to generate an equivalent trajectory ensemble. The neural network models are integrated numerically using a relative error tolerance of 10^{-3} and an absolute error tolerance of 10^{-6}.

IV. CONCLUSIONS

This work presented two novel neural ODE-based architectures for modeling the dynamics of memristor devices. The proposed models leverage the power of neural networks to capture complex, nonlinear behaviors inherent in memristive systems, while maintaining the interpretability and physical consistency afforded by the ODE framework. The proposed models were applied for modeling of real Self-Directed Channel (SDC) memristors. Extensive evaluation against experimental data demonstrates that the neural ODE approach significantly outperforms traditional phenomenological models, such as the MMS and GMMS models, in terms of accuracy and fidelity. The results highlight the capability of the neural ODE approach to learn intricate device dynamics directly from data, thereby providing a powerful tool for simulating and understanding memristor behavior. Hyperparameter sensitivity analysis further elucidates the critical factors influencing model performance, guiding future optimization efforts. Obtained results show that the computation time needed to optimize the proposed models is smaller than that for existing models. On the other hand the evaluation time needed to compute trajectories using the proposed models is significantly larger than for the MMS and GMMS models.

The proposed models may also be used for modeling of memristors produced in other technologies. The findings of this study underscore the potential of neural ODEs as a versatile and effective framework for modeling complex physical systems, with applications in the design and analysis of memristor-based circuits. Developed methods pave the way to more accurate and efficient simulation tools, facilitating advancements in memristor technology and its integration into emerging computing paradigms, and open avenues for future research in data-driven modeling of nonlinear dynamical systems.

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